- Differentiation : The special type of any existing limit, $\lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$ is called the derivative of the function f with respect to x and is denoted by $f'(x)$. If y is a function of x, and has a derivative, then the differential coefficient of y with respect to x is denoted by $\frac{dy}{dx}$.

- Some rules on differentiation:

(i) Derivative of a constant function is zero. $f'(c) = 0$, where c is some constant.

(ii) If u is a function of x and k is some constant and *dash* denotes the differentiation, $[k\,u]' = k\,[\,u\,]'$

(iii) $(u \pm v)' = u' \pm v'$

(iv) $(u\,v)' = u'v + u\,v'$

(v) $\left[\frac{u}{v}\right]' = \frac{u'v - uv'}{v^2}$

- Important formulae:

(i) $(x^n)' = n\,x^{n-1}$

(ii) $(e^x)' = e^x$

(iii) $(\log x)' = \frac{1}{x}$

- Integration

Integration is the reverse process of differentiation. It is also called anti-derivative.

Rules and some formulae on integration:

(i) $\int k\,dx = kx$

(ii) $\int x^n dx = \frac{x^{n+1}}{n+1}$

(iii) $\int e^x dx = e^x$

(iv) $\int \frac{1}{x} dx = \log x$

(v) $\int (u \pm v)\,dx = \int u dx \pm \int v dx$

- Definite Integral : $\int_a^b f(x)\,dx = F(b) - F(a)$

- Double integrals : The double integral defined on the function $f(x,y)$ is denoted as $\int_a^b \int_c^d f(x,y)\,dy\,dx$. The function $f(x,y)$ is integrated with respect to y first by treating $f(x)$ as constant and then integrate with respect to x and apply limits of x and simplify we get the value of integral.